Prediction and Analysis of Ambient Air Quality in Harbin Based on Time Series Analysis Model

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***Abstract*—To grasp the future trend of ambient air quality more accurately and provide more reliable data support for formulating and implementing environmental protection departments' policies. This paper establishes the Winters model and mean GM (1,1) model to predict the ambient air quality in the next five years, based on Harbin's atmospheric environmental quality monitoring data from 2015 to 2021. The results show that the seasonal characteristics of ambient air quality in Harbin are still not eliminated, but the overall trend is improving year by year. Using the combined model can make long-term predictions for smaller time units, enhancing work precision and short-term predictions for larger time units. Furthermore, it makes the comprehensive study and judgment of the future change trend of environmental factors more reasonable and the prediction results more meaningful.**

Keywords—ambient air quality, time series analysis model, prediction

I. Introduction

The quality of China's eco-environment has generally improved in recent years. During the "13th Five Year Plan" period, the number of cities meeting national air quality standard and the average percentage of days meeting air quality standard increased, while the proportion of days with heavy pollution and the concentration of significant pollutants decreased. In addition, for environmental protection, the 14th Five-Year Plan puts forward higher requirements [1]. Therefore, more advanced research methods and more accurate prediction models are required to accurately predict the future development and change trend of ambient air quality. Ambient air quality prediction refers to the judgment made on the future change trend of air quality [2]. It is based on historical data to infer the present and future. Moreover, accurate prediction of the future air quality trend can provide more precise goals and directions for formulating environmental policies in the next quarter or year.

The classic prediction methods are primarily annual predictions with a single gray model method, which is good at short-term prediction and can reliably anticipate the future change trend of a particular batch of specific data. However, the disadvantages are also obvious. First, the prediction will be somewhat speculative and subjective due to the limitation of specific data, so once a batch of data is changed, it needs to be re-predicted. Second, a single model cannot make accurate long-term prediction, and most of the predictions are based on annual average data. Therefore, in order to improve the precision of environmental management, the monthly data should be used as the prediction unit. Therefore, in this paper, firstly, the grey model was used to make short-term annual prediction, providing data support for customizing annual strategic policies in the next five years. Secondly, the time series analysis model was adopted to make long-term monthly prediction, which can better grasp the changing air quality trend in the next year.

II. Data Source and Methodology

A. Data Source

The data is derived from Harbin National Controlled Ambient Air Automatic Monitoring Station and Harbin Eco-Environmental Monitoring Centre. There are 12 automatic monitoring points in Harbin, the levels of fine particles ($PM_{2.5}$), inhalable particles (PM_{10}), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO) and ozone (O_3) were monitored continuously for 24 hours per day. Levels of the above six air pollutants in Harbin were collected from January 2015 to December 2021[3-4].

B. Prediction Model

1) Introduction and modeling steps of grey GM (1,1) model: The GM (1,1) model has four basic forms, they are the mean GM (1,1) model, the original differential GM (1,1) model, the mean differential GM (1,1) model and the discrete GM (1,1) model [5]. Among those, the original differential GM (1,1) model, the mean differential GM (1,1) model and the discrete GM (1,1) model can accurately simulate the homogeneous index sequence, and the mean GM (1,1) model is suitable for simulating the non-exponential growth sequence and the shock sequence. Therefore, this paper selects the mean GM (1,1) model to model the data. The modeling steps are as follows:

a) Set X_0 as a non-negative sequence: $X_0 = (x_0(1), x_0(2), \cdots, x_0(n))$.

b) Generate 1-AGO (that is, one time accumulation) sequence: X_1 : $X_1 = (x_1(1), x_1(2), \cdots, x_1(n))$, where $x_1(k) = \sum_{i=1}^{k} x_0(i); k = 1, 2, \cdots, n$.

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c) Generate the nearest mean sequence Z_1 : $Z_1 = (z_1(2), z_1(3), \cdots, z_1(n))$, where $z_1(k) = 0.5(x_1(k) + x_1(k-1)), k = 1, 2, \cdots, n$.

d) Solving GM (1,1) model $x_0(k) + az_1(k) = b$ by the least square method: If $\hat{a} = [a, b]^T$ is parameter column, and $Y = \begin{bmatrix} x_0(2) \\ x_0(3) \\ \vdots \\ x_0(n) \end{bmatrix}$, $B = \begin{bmatrix} -Z_1(2), 1 \\ -Z_1(3), 1 \\ \vdots \quad \vdots \\ -Z_1(n), 1 \end{bmatrix}$, the solution of the parameter is $\hat{a} = [a, b]^T = (B^T B)^{-1} B^T Y$, Where a is called development coefficient and b is grey action quantity [6-7].

e) Substitute the estimated parameters into the model to obtain the time response function sequence: $\hat{x}_1(k+1) = (x_0(1) - \frac{b}{a})e^{-ak} + \frac{b}{a}, k = 1, 2, \cdots, n$.

f) Verify the validity of the model: When the development coefficient of GM (1,1) model $|a| \geq 2$, GM (1,1) model is meaningless, that is, when $|a| < 2$, GM (1,1) model is meaningful.

g) Model accuracy analysis: After the model validity test is passed, the prediction model needs to be tested for accuracy. Standard precision tests include average relative error and correlation degree (that is, the calculation result of the correlation degree is less than the given threshold of the correlation degree grade, which is called the correlation degree qualified model), and the ratio of mean square error (that is, the calculated result of mean square error ratio is less than the given threshold of mean square error ratio, which is called the qualified model of mean square error ratio), and the small error probability (that is, the calculated result of small error probability is less than the given threshold of small error probability, which is called the qualified model of small error probability) [8-9]. If the accuracy test result is low, the prediction result is meaningless. In this paper, the average relative error is selected to test the accuracy of the model. If the accuracy test result is good, it can be predicted according to the time response function sequence. Refer to Table I for the accuracy level.

TABLE I. REFERENCE TABLE OF ACCURACY INSPECTION GRADE

Accuracy level	Average Relative Error	Relevancy	Mean Variance Ratio	Small Error Probability
One-level	0.01	0.90	0.35	0.95
Two-level	0.05	0.80	0.50	0.80
Three-level	0.10	0.70	0.65	0.70
Four-level	0.20	0.60	0.80	0.60

2) Introduction and modeling steps of time series analysis model: The commonly used time series analysis models for non-stationary time series include the ARIMA(Autoregressive Integrated Moving Average) model, seasonal model, residual autoregression model, conditional heteroscedasticity model, etc. This paper selects Holt Winters prediction model [10-12] (including simple seasonal model). This model has good prediction results for time series with seasonality and trend, that is, in each cycle, three weights of one horizontal component, one trend component, and one seasonal component are used to update the components. The initial values of the horizontal and trend components are obtained by linear regression of time. The initial value of the seasonal component is obtained by regression of the dummy variable of data after trend removal. When the horizontal component is multiplied by the seasonal component, it is the multiplication model; when the horizontal component and the seasonal component are added, it is the addition model[13]. The model and prediction equation are as follows:

$$\begin{cases} L_t = \alpha(Y_t - S_{t-s}) + (1-\alpha)(L_{t-1} + T_{t-1}) \\ T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1} \\ S_t = \gamma(Y_t - L_t) + (1-\gamma)S_{t-s} \\ Y_t = L_{t-1} + T_{t-1} + S_{t-s} \end{cases}$$

$$\begin{cases} L_t = \alpha(Y_t / S_{t-s}) + (1-\alpha)(L_{t-1} + T_{t-1}) \\ T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1} \\ S_t = \gamma(Y_t / L_t) + (1-\gamma)S_{t-s} \\ Y_t = (L_{t-1} + T_{t-1})S_{t-s} \end{cases}$$

In the above formula, L is horizontal effect, T is trend effect, S is seasonal effect, α is smoothing coefficient of horizontal effect, β is smoothing coefficient of trend effect, γ is smoothing coefficient of seasonal effect. The value range of the three smoothing coefficients is $[0,1]$, t is period, s is seasonal cycle (Quarterly data $s = 4$, monthly data $s = 12$).

The modeling steps are as follows:

a) According to the time sequence diagram, the time sequence is determined to be a non-white noise sequence.

b) Calculate the autocorrelation and partial autocorrelation *coefficients*, and select an appropriate model for fitting.

c) Estimate the values of unknown parameters in the model.

d) Check the validity of the model. If the fitted model does not pass the inspection, turn to step *b)* and re-select the model for re-fitting.

e) Predict the future trend according to the prediction *equation* [14].

III. EXAMPLE ANALYSIS OF AMBIENT AIR QUALITY PREDICTION IN HARBIN

A. Descriptive Statistical Analysis

According to the ambient air quality data of Harbin from 2015 to 2021, descriptive statistical analysis was conducted on the concentrations of 6 pollutants [15]. Among them, the measurement of centralized trend calculates the average, maximum, minimum, and quantile; The measure of dispersion calculates variance and standard deviation [16]. The descriptive statistical results are shown in Table II.

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TABLE II. DESCRIPTIVE STATISTICAL ANALYSIS OF POLLUTANT CONCENTRATION FROM JANUARY 2015 TO DECEMBER 2021

Indices	Concentrated Trend						Degree of Dispersion	
	ave	*min*	*max*	*25%*	*50%*	*75%*	*variance*	*SD*
Fine particles (μg /m³)	49.4	12.0	155.0	22.0	37.5	65.8	1250.5	35.4
Inhalable particles (μg /m³)	74.0	26.0	185.0	43.5	62.0	94.8	1403.6	37.5
Nitrogen dioxide (μg /m³)	38.8	22.0	77.0	30.0	35.0	45.0	169.9	13.0
Sulfur dioxide (μg /m³)	23.6	6.0	157.0	9.3	16.0	30.0	540.5	23.2
Carbon monoxide (mg/m³)	1.2	0.5	3.6	0.8	1.1	1.6	0.4	0.6
Ozone (μg /m³)	98.4	37.0	173.0	62.3	94.0	122.0	1383.1	37.2

Fig. 1 shows the monthly changes of six pollutants from January 2015 to December 2021. It can be seen from Fig. 1 that the seasonal and trend changes of the six pollutants are obvious. Among them, fine particles, inhalable particles, nitrogen dioxide and sulfur dioxide all present a "U-shaped" change law of first falling and then rising. The wave peaks mainly appear in winter, and the wave troughs mainly appear in summer, with a downward trend year by year. The ozone presents an "inverted U-shaped" change law of first rising and then falling. The wave crest mainly appears in summer, and the wave trough mainly appears in winter, and presents a downward trend year by year.

B. Model Establishment and Solution

1) Establishment and solution of time series analysis model: IBM SPSS 26.0 statistical analysis software was used to establish, fit, and predict the data model. The data were modeled, and the significance level α was set as 0.05. Define the date and time for the monthly data of six air pollutants from January 2015 to December 2021 and generate the monthly time series; The time series prediction module in the software is used to create the traditional time series model. Check the expert modeler option (the expert modeler can automatically select the model for modeling according to the data). According to the data characteristics of the six air pollutant indicators, the modeling results automatically select the Holt Winters model and the simple seasonal model.

From the fitting results of the model, the stable R^2 of the six indicators is between 0.63-0.77, R^2 is between 0.67-0.86, and the average absolute percentage error is between 0.2% and 13.9% (the average absolute percentage errors of nitrogen dioxide, sulfur dioxide, and carbon monoxide are all less than 5.00%). In general, the fitting effect of the model is perfect. Refer to Table III for the statistical table of the model fitting degree.

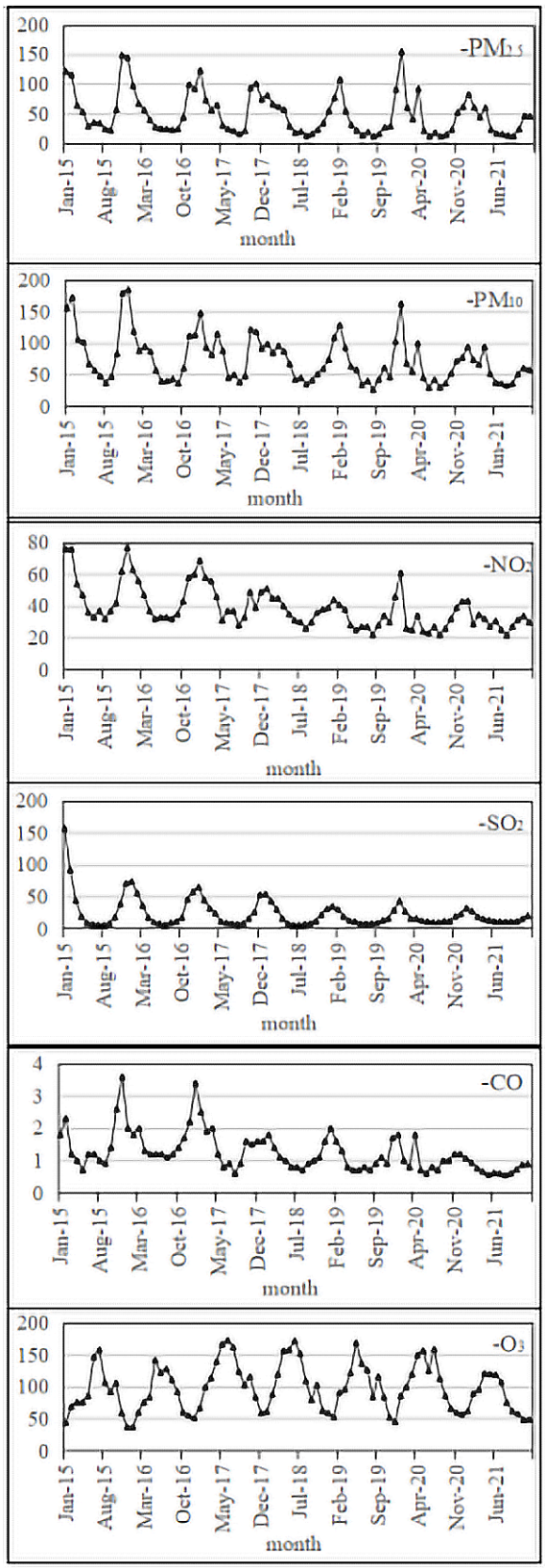

Fig. 1. Monthly Change Trend of Concentration of Six Pollutants.

TABLE III. STATISTICS OF TIME SERIES ANALYSIS MODEL PARAMETERS

Index	Model Fit Statistics			Ljung Box Statistics		
	Stable R^2	*R^2*	*MAE (%)*	*Statistics*	*DF*	*Significance*
$PM_{2.5}$	0.66	0.72	11.6	25.58	15	0.043
PM_{10}	0.69	0.72	13.9	23.88	15	0.067
NO_2	0.63	0.82	4.1	19.51	15	0.191

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SO_2	0.77	0.86	3.2	37.12	15	0.001
CO (per95)	0.72	0.67	0.2	17.46	15	0.292
O_3 (per90)	0.71	0.82	12.5	30.49	16	0.016

2) Establishment and solution of mean GM (1,1) model: Based on the annual data of six air pollutants from 2015 to 2021, using the mean GM (1,1) model of the grey prediction model module in the grey modeling software V6.0 to establish, fit and forecast the data.

From the fitting results of the model, the average relative errors of the six pollutant indicators are from 4.1% to 9.7%, all less than 10%. In general, the fitting effect of the model is good. Refer to the statistical Table IV for the model fitting degree.

TABLE IV. STATISTICAL TABLE OF GREY GM (1,1) MODEL PARAMETERS

Indices	$PM_{2.5}$	PM_{10}	NO_2	SO_2	CO (per95)	O_3 (per90)
Average Relative Error (%)	9.7	6.2	4.1	6.1	7.9	7.6

C. *Prediction and result analysis of ambient air quality in Harbin in 2022 and the next five years*

The data of six air pollutants are extrapolated through the model to obtain the prediction data from January to December 2022 and 2022 to 2025. Refer to Table V for the prediction results.

TABLE V. PREDICTION DATA OF SIX AIR POLLUTANTS FROM JANUARY TO DECEMBER 2022 AND 2022-2025

μg/m³

Predict Time	$PM_{2.5}$	PM_{10}	NO_2	SO_2	CO (per95)	O_3 (per90)
January 2022	73	88	40	30	1	48
February 2022	54	71	32	23	0.9	75
March 2022	37	59	29	15	0.7	89
April 2022	39	64	26	14	0.6	111
May 2022	17	42	20	12	0.4	132
June 2022	14	28	21	10	0.4	131
July 2022	15	29	21	10	0.5	131
August 2022	11	24	18	10	0.4	92
September 2022	12	28	21	11	0.4	79
October 2022	27	46	25	11	0.6	74
November 2022	48	64	28	16	0.7	49
December 2022	53	68	32	20	0.9	40
2023	34	52	25	11	1	132
2024	32	49	23	10	0.9	134
2025	30	46	21	8	0.8	136

Fig. 2 to Fig.7 reflect the changing trend of the predicted values of six air pollutants, with the horizontal axis representing time and the vertical axis representing concentration (in which the measure of CO is mg/m³ and the other five measures are μg/m³).

1) $PM_{2.5}$: As shown in Fig. 2, the $PM_{2.5}$ concentration showed an overall decreasing trend from an annual perspective; From a monthly perspective, it offers a trend of decrease first and then increase, with a minimum value of 11 μg /m³, appeared in August, the maximum value is 73 μg/m³, appeared in January.

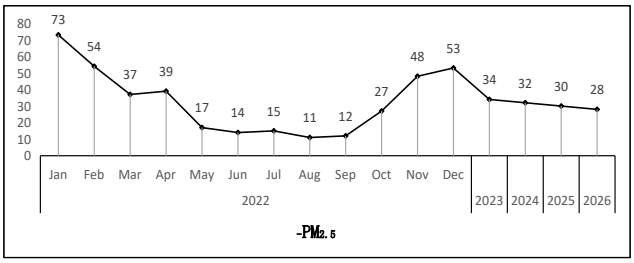

Fig. 2. Change Trend of Predicted Values of $PM_{2.5}$.

2) PM_{10}: As shown in Fig. 3, the PM_{10} concentration of inhalable particulates showed an overall decreasing trend from an annual perspective; From a monthly perspective, it offers a trend of decline first and then rise, with a minimum value of 24 μg/m³, appeared in August, the maximum value is 88 μg/m³, appeared in January.

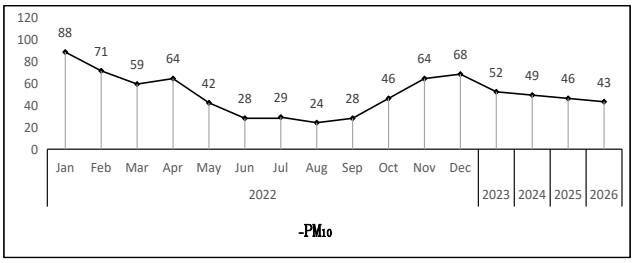

Fig. 3. Change Trend of Predicted Concentration of PM_{10}.

3) NO_2: As shown in Fig.4, the NO_2 concentration showed an overall decreasing trend from an annual perspective; From a monthly perspective, it offers a trend of decreasing first and then increasing, with a minimum value of 18 μg/m³, appeared in August, the maximum value is 40 μg/m³, appeared in January.

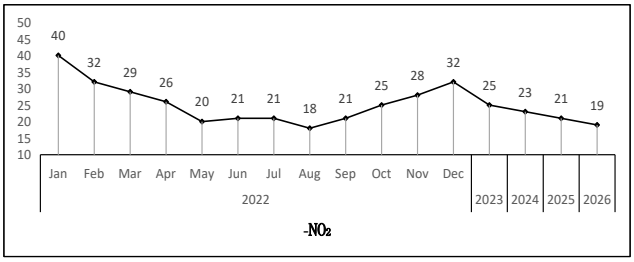

Fig. 4. Change Trend of Predicted Values of NO_2.

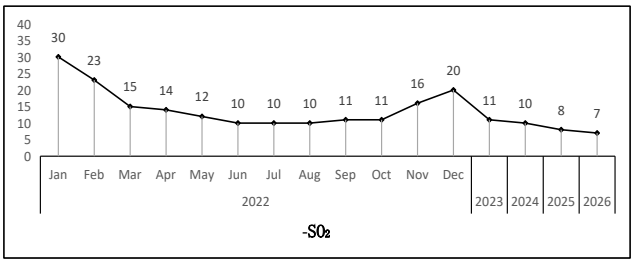

Fig. 5. Change Trend of Predicted SO_2.

4) SO_2: As shown in Fig. 5, the SO_2 concentration showed an overall decreasing trend from an annual perspective; From a monthly perspective, it offers a trend of decreasing first and then increasing, with a minimum value of 10 μg/m³, appeared from June to August, the maximum value is 30 μg/m³, appeared in January.

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5) CO: As shown in Fig. 6, the CO concentration showed an overall decreasing trend from an annual perspective; From a monthly perspective, it offers a trend of decreasing first and then increasing, with the minimum value of 0.4 mg/m^3, appeared from May to September, and the maximum value of 1.0 mg/m^3, appeared in January.

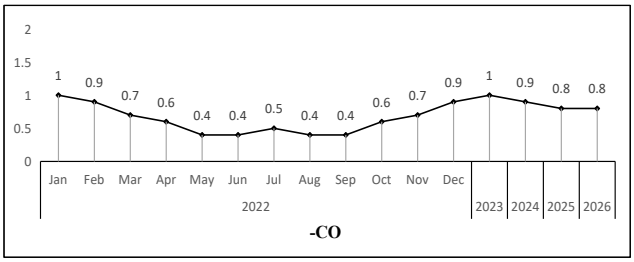

Fig. 6. Change Trend of Predicted CO.

6) O_3: As shown in Fig. 7, from an annual perspective, the O_3 concentration shows a trend of overall decline; From a monthly perspective, it offers a trend of increasing first and then decreasing. The minimum value is 40 mg/m^3, which appears in December, and the maximum value is 132 mg/m^3, which appears in May.

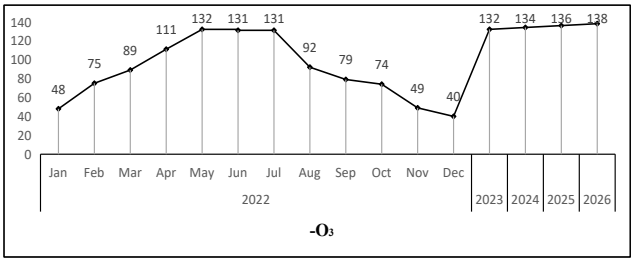

Fig. 7. Change Trend of Predicted O_3.

IV. Conclusion

Based on the monthly and annual levels of six air pollutants in Harbin, this paper forecasts the ambient air quality of Harbin in 2022 and the next five years through descriptive analysis, mean GM(1,1) model analysis, and Winters model analysis. The fitting degree and prediction accuracy of the model are both high. The Winters model has a good prediction effect on environmental elements with periodicity and seasonality. The grey GM model is suitable for short-term prediction. Using the combined model can make not only long-term predictions for smaller time units but also short-term predictions. Therefore, it is more reasonable to comprehensively study and judge the future change trend of environmental elements.

The monthly forecast results show that the seasonality of the ambient air quality in Harbin is still not eliminated, showing a "U" distribution. Among them, the air quality from January to March and November to December is relatively poor, mainly due to coal combustion during the heating period. In recent years, the government and other relevant environmental protection departments have taken measures to minimize the impact of heating enterprises on ambient air quality during the heating period, such as staggered starting of coal-fired heating boilers, staggered production of cement, prohibition of burning straw, and dust control. When viewed through the lens of the annual forecast results, the overall trend is improving year by year, indicating that the government work has been successful. Therefore, it is necessary to predict the ambient air quality in Harbin in this paper, which can provide more essential data support for the government departments to formulate comprehensive plans for future environmental protection-related policies.

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